

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Assignment 2: Tetris Local Search

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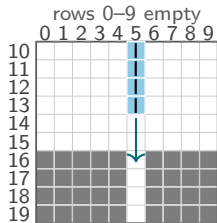
September 10, 2026

Tetris Basics

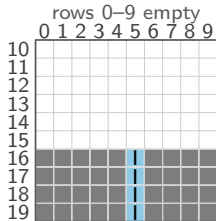
Play Tetris

- ▶ Move and rotate each falling tetromino to fit it into the stack.
- ▶ Fill every cell in a horizontal row to clear that row and earn points.
- ▶ Keep the stack below the top; pieces fall faster as levels increase.

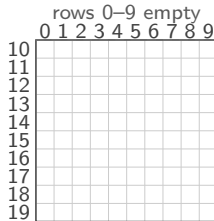
Falling I



Four filled rows

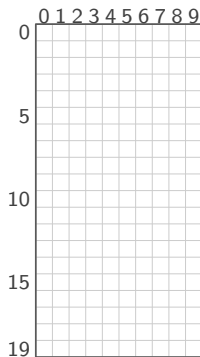


Rows cleared



A Fixed Planning Problem

Initial board

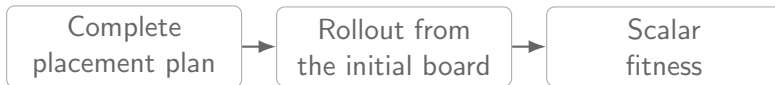


10 columns $\times$ 20 rows

Known sequence: 12 pieces

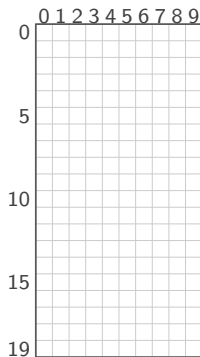
O I T L J S Z I T O L J

One objective evaluation



A Fixed Planning Problem

Initial board

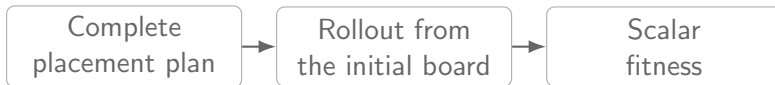


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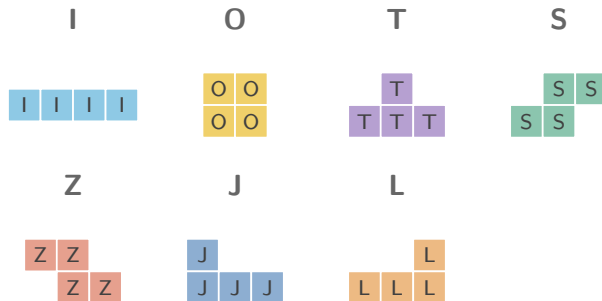
O I T L J S Z I T O L J

One objective evaluation



- Search changes a complete plan.
- A rollout executes that plan, one piece at a time.

The Seven Tetrominoes



Every tetromino has four cells. The pictured orientation has rotation index 0.

Unique Clockwise Rotations

Piece	Unique orientations, in index order	Count
O	 0	1
I	 0 →  1	2
T	 0 →  1 →  2 →  3	4

Unique Clockwise Rotations

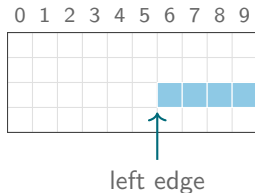
Piece	Unique orientations, in index order	Count
O	 0	1
I	 0 →  1	2
T	 0 →  1 →  2 →  3	4

- ▶ Rotate clockwise, normalize to the upper-left corner, and omit duplicates.
- ▶ The square has only rotation 0; its turned copies add no new orientation.

Left-Edge Columns and Horizontal Fit

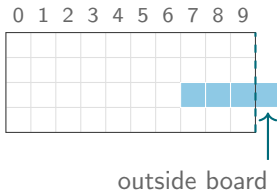
A placement is (rotation index, left column).

Horizontal I: (0, 6)



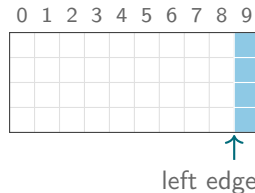
Valid: columns 6–9

Horizontal I: (0, 7)



Invalid: includes column 10

Vertical I: (1, 9)



Valid: only column 9

For rotated width w , the left column ranges from 0 through $10 - w$.

Shape-Dependent Placement Domains

Piece	Rotation widths	Positions per rotation	Domain size
O	2	9	9
I	4, 1	$7 + 10$	17
T	3, 2, 3, 2	$8 + 9 + 8 + 9$	34
S, Z	3, 2	$8 + 9$	17 each
J, L	3, 2, 3, 2	$8 + 9 + 8 + 9$	34 each

Every domain entry is a horizontally valid (rotation, column) pair.

Shape-Dependent Placement Domains

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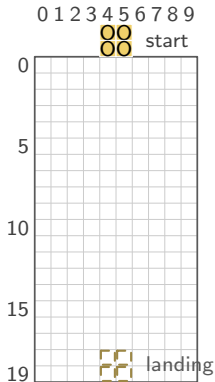
Every domain entry is a horizontally valid (rotation, column) pair.

The domain depends on the piece shape.

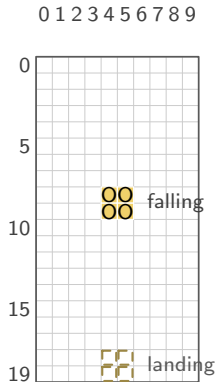
The current board affects the rollout and score, including possible top-out.

A Vertical Hard Drop

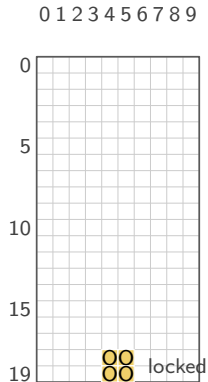
Start above the board



Vertical descent



Lock when blocked



The placement fixes rotation and column; the board determines the landing row.

A Four-Line Clear

The supplied well has one empty column in each of its bottom four rows.

Vertical I at (1, 5)

rows 0–14 empty

	0	1	2	3	4	5	6	7	8	9
15										
16										
17										
18										
19										

Four completed rows

rows 0–14 empty

	0	1	2	3	4	5	6	7	8	9
15										
16						I				
17						I				
18						I				
19						I				

After clearing

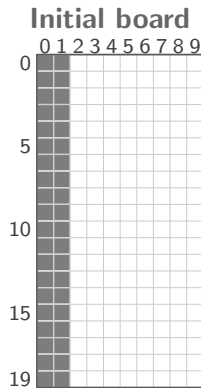
rows 0–14 empty

	0	1	2	3	4	5	6	7	8	9
15										
16										
17										
18										
19										

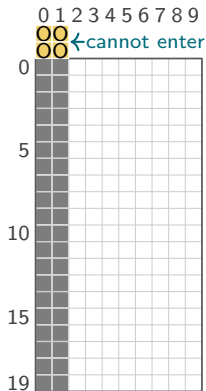
The engine clears every full row after the lock.

This first placement adds one placed piece and four cleared lines.

Top-Out Above the Board



Attempted O at (0,0)



A horizontally valid placement

- ▶ The O fits columns 0 and 1.
- ▶ It cannot lock wholly inside the board.
- ▶ Top-out ends this rollout.

The failed piece is not added.

The board stays unchanged, and later placements are not executed.

A Complete Placement Plan

Three-piece example: O, I, T; $N = 3$.

Plan A: one placement for each piece

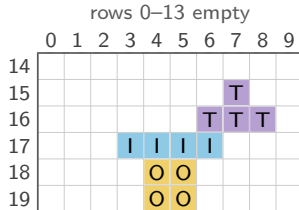
Piece 0	Piece 1	Piece 2
O (0, 4)	I (0, 3)	T (0, 6)

$$A = ((0, 4), (0, 3), (0, 6))$$

Each pair belongs to that piece's domain.

All three pairs are chosen before evaluation.

After evaluating A



O, then I, then T

Neighbors by Changing One Placement

Three-piece example: O, I, T; $N = 3$.

Current A	O (0, 4)	I (0, 3)	T (0, 6)	
Neighbor B	O (0, 0)	I (0, 3)	T (0, 6)	Change O
Neighbor C	O (0, 4)	I (1, 9)	T (0, 6)	Change I
Neighbor D	O (0, 4)	I (0, 3)	T (2, 0)	Change T

Neighbors by Changing One Placement

Three-piece example: O, I, T; $N = 3$.

Current A	O (0, 4)	I (0, 3)	T (0, 6)	
Neighbor B	O (0, 0)	I (0, 3)	T (0, 6)	Change O
Neighbor C	O (0, 4)	I (1, 9)	T (0, 6)	Change I
Neighbor D	O (0, 4)	I (0, 3)	T (2, 0)	Change T

Exactly one placement changes to another value in its domain.

A neighbor may move a piece several columns across the board.

Changing One Placement or Two

Three-piece example: O, I, T; $N = 3$.

Current A

O (0, 4)

I (0, 3)

T (0, 6)

One changed placement

O (0, 4)

I (1, 9)

T (0, 6)

I changes $(0, 3) \rightarrow (1, 9)$.

Both rotation and column can change together.

This is a neighbor of A.

Changing One Placement or Two

Three-piece example: O, I, T; $N = 3$.

Current A



One changed placement

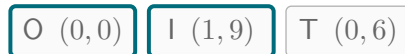


I changes $(0, 3) \rightarrow (1, 9)$.

Both rotation and column can change together.

This is a neighbor of A.

Two changed placements



Both O's and I's placements change.

Each pair is valid for its piece.

This is a valid plan, but not a neighbor of A.

Neighborhood Size and Canonical Order

Three-piece example: O, I, T; $N = 3$.

Neighbors of one plan

$$(9 - 1) + (17 - 1) + (34 - 1) = 57$$

Choose one piece and change its placement.

The entire candidate space

$$9 \times 17 \times 34 = 5,202$$

Choose a placement for every piece.

Canonical neighbor order



For each piece: rotation index first, then increasing column.

Skip the current placement; keep every other placement unchanged.

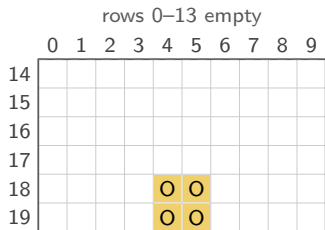
An Early Edit and Later Landings

Three-piece example: O, I, T; $N = 3$.

O → I → T After placing O

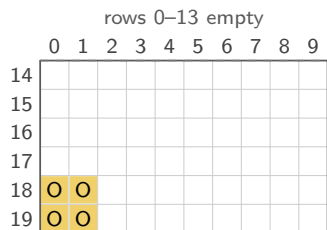
Plan A

O (0, 4) I (0, 3) T (0, 6)



Plan B

O (0, 0) I (0, 3) T (0, 6)



I and T keep their placements; changing O changes their landings.

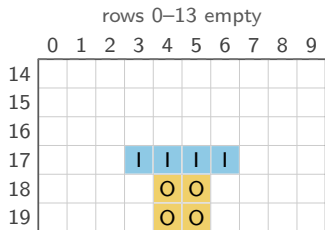
An Early Edit and Later Landings

Three-piece example: O, I, T; $N = 3$.

O $\rightarrow$ I $\rightarrow$ T After placing I

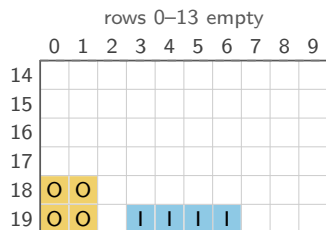
Plan A

O (0, 4) I (0, 3) T (0, 6)



Plan B

O (0, 0) I (0, 3) T (0, 6)



I and T keep their placements; changing O changes their landings.

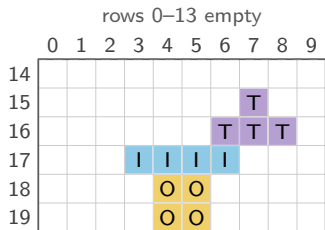
An Early Edit and Later Landings

Three-piece example: O, I, T; $N = 3$.

O $\rightarrow$ I $\rightarrow$ T After placing T

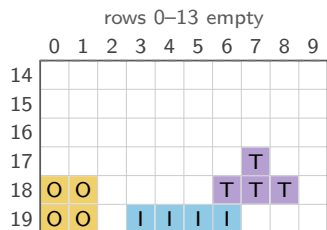
Plan A

O (0, 4) I (0, 3) T (0, 6)



Plan B

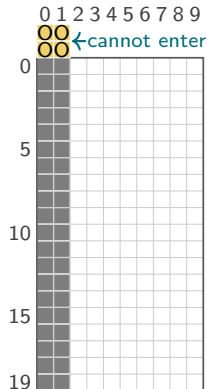
O (0, 0) I (0, 3) T (0, 6)



I and T keep their placements; changing O changes their landings.

Unused Placements and Fitness Plateaus

Three-piece example: O, I, T; $N = 3$.



Two complete plans for O, I, T

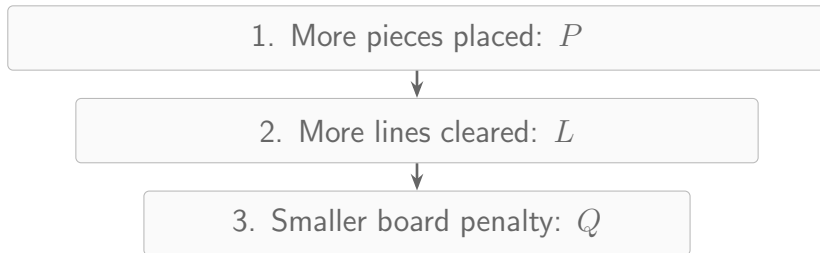
Plan U	O (0,0)	I (0,0)	T (0,0)
Plan V	O (0,0)	I (1,9)	T (0,0)

- ▶ Both rollouts top out on the first O.
- ▶ The I placement differs; neither version is executed.
- ▶ Both final boards and fitness values are identical.

Distinct neighboring plans can share one score.

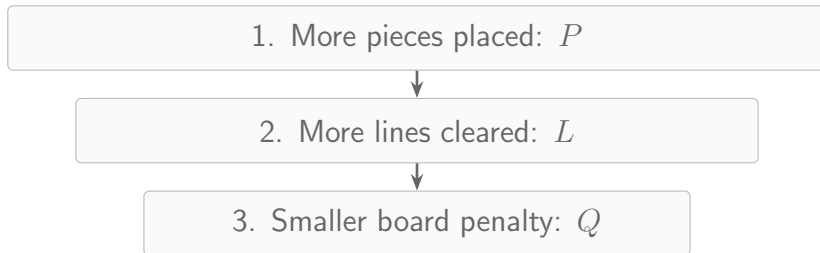
Fitness Priorities

Every complete plan receives one score; higher is better.



Fitness Priorities

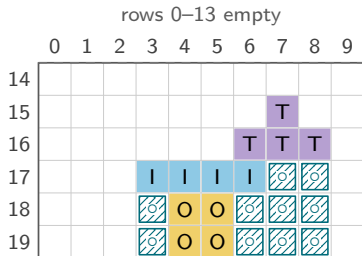
Every complete plan receives one score; higher is better.



Compare the next criterion only when the earlier ones tie.

Holes Below Occupied Cells

Three-piece example: O, I, T; $N = 3$.



Plan A after all three drops

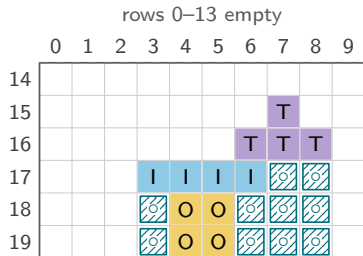
Hole definition

An empty cell with an occupied cell somewhere above it in the same column.

- ▶ Count cells, including several in one column.
- ▶ Open cells above the stack are not holes.

Holes Below Occupied Cells

Three-piece example: O, I, T; $N = 3$.



Plan A after all three drops

Hole definition

An empty cell with an occupied cell somewhere above it in the same column.

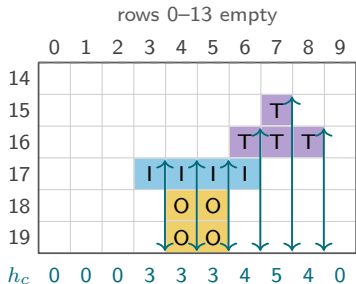
- ▶ Count cells, including several in one column.
- ▶ Open cells above the stack are not holes.

$$\text{holes} = 2 + 2 + 3 + 3 = 10.$$

$$\text{Hole penalty} \quad 5 \times 10 = 50$$

Aggregate Height

Three-piece example: O, I, T; $N = 3$.



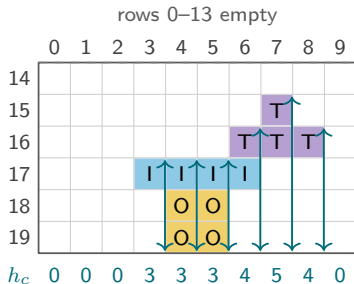
Column height

Measure from the floor to the highest occupied cell. An empty column has height zero.

Holes inside that span still contribute to height.

Aggregate Height

Three-piece example: O, I, T; $N = 3$.



Column height

Measure from the floor to the highest occupied cell. An empty column has height zero.

Holes inside that span still contribute to height.

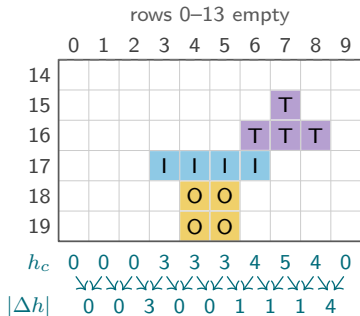
$$H = \sum_{c=0}^9 h_c$$

$$H = 3 + 3 + 3 + 4 + 5 + 4 = 22.$$

The ten heights appear below their columns.

Bumpiness Between Columns

Three-piece example: O, I, T; $N = 3$.



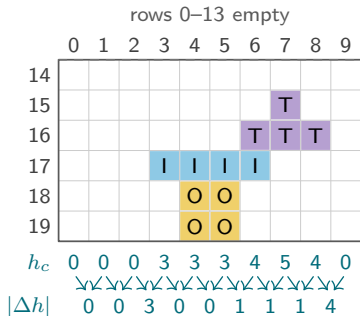
Adjacent height differences

Take the absolute difference for each of the nine neighboring column pairs.

$$B = \sum_{c=0}^8 |h_c - h_{c+1}|.$$

Bumpiness Between Columns

Three-piece example: O, I, T; $N = 3$.



Adjacent height differences

Take the absolute difference for each of the nine neighboring column pairs.

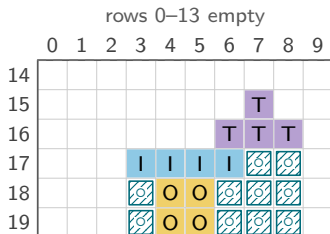
$$B = \sum_{c=0}^8 |h_c - h_{c+1}|.$$

$$B = 3 + 1 + 1 + 1 + 4 = 10.$$

The other four differences are zero. Include empty edge columns; do not compare with an outside wall.

The Board Penalty

Three-piece example: O, I, T; $N = 3$.

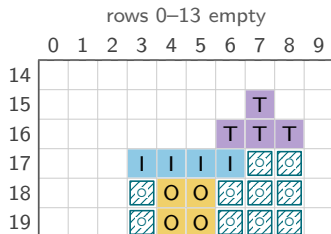


Count the final board, after line clearing.

Metric	Count	Contribution
Holes	10	$5 \times 10 = 50$
Aggregate height	22	22
Bumpiness	10	10

The Board Penalty

Three-piece example: O, I, T; $N = 3$.



Count the final board, after line clearing.

Metric	Count	Contribution
Holes	10	$5 \times 10 = 50$
Aggregate height	22	22
Bumpiness	10	10

$$Q = 5 \text{ holes} + H + B$$

$$Q = 50 + 22 + 10 = 82.$$

Lower Q is better when pieces placed and lines cleared tie.

Designing the Fitness

$$F = P + \frac{L - Q/1381}{4N + 1}, \quad N = \text{number of planned placements.}$$

- Choose the penalty** $Q = 5 \text{ holes} + H + B \leq 5(10 \cdot 20) + 10 \cdot 20 + 9 \cdot 20 = 1380$.
The weight 5 is a heuristic choice; 1380 is a conservative bound.
- Prioritize lines** $0 \leq Q/1381 < 1$, so $L - Q/1381 \in (L - 1, L]$.
With P fixed, an extra line always wins; lower Q breaks ties.
- Prioritize pieces** At most four lines per piece: $-1 < L - Q/1381 \leq 4N$.
Dividing by $4N + 1$ makes its variation < 1 ; an extra piece always wins.

From Metrics to Fitness

Three-piece example: O, I, T; $N = 3$.

$$F = P + \frac{L}{4N + 1} - \frac{Q}{(4N + 1) 1381}.$$

Placed $P = 3$

Cleared $L = 0$

Penalty $Q = 82$

From Metrics to Fitness

Three-piece example: O, I, T; $N = 3$.

$$F = P + \frac{L}{4N + 1} - \frac{Q}{(4N + 1) 1381}.$$

Placed $P = 3$

Cleared $L = 0$

Penalty $Q = 82$

$$F(A) = 3 + \frac{0}{13} - \frac{82}{13 \times 1381} \approx 2.995433.$$

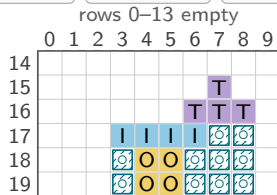
The planned sequence length

N counts all planned placements, including unused ones after top-out. Here $N = 3$; assignment levels use $N = 12$, so $4N + 1 = 49$.

Two Neighboring Plans

Three-piece example: O, I, T; $N = 3$.

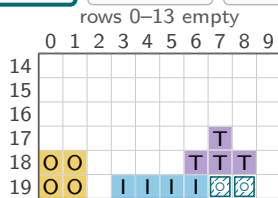
O (0, 4) I (0, 3) T (0, 6)



Plan A: holes 10, height 22, bumpiness 10

$$Q(A) = 82, \quad F(A) \approx 2.995433.$$

O (0, 0) I (0, 3) T (0, 6)



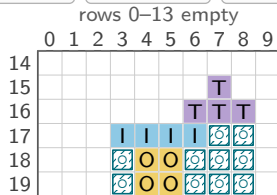
Plan B: holes 2, height 14, bumpiness 8

$$Q(B) = 32, \quad F(B) \approx 2.998218.$$

Two Neighboring Plans

Three-piece example: O, I, T; $N = 3$.

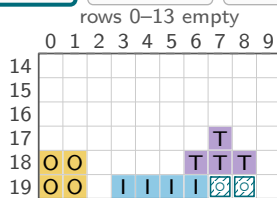
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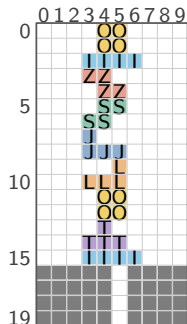
Plan B: holes 2, height 14, bumpiness 8

$$Q(B) = 32, \quad F(B) \approx 2.998218.$$

Both place three pieces and clear zero lines. Plan B wins on board quality.

The Twelve-Piece Well Level

Supplied well.json: $N = 12$, so $4N + 1 = 49$.

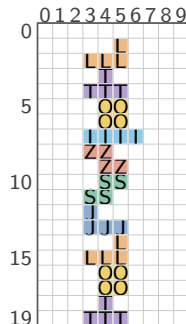


Original plan

First placement: $I(0,3) \rightarrow I(1,5)$

The other eleven placements stay fixed.

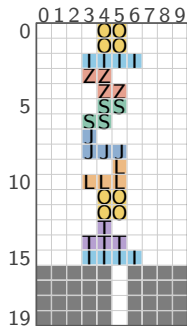
	Original	Neighbor
Pieces placed	9	11
Lines cleared	0	4
Board penalty	272	246
Fitness	8.995980	11.077997



Neighboring plan

The Twelve-Piece Well Level

Supplied well.json: $N = 12$, so $4N + 1 = 49$.



Original plan

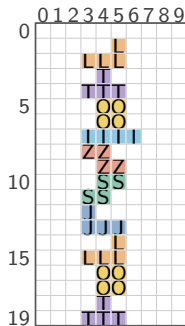
First placement: I (0, 3) → I (1, 5)

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	Original	Neighbor
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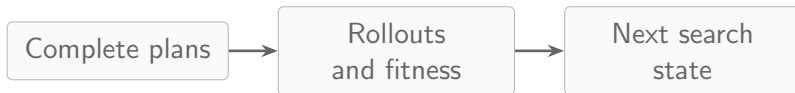
Four rows clear on the first drop.

Both complete rollouts eventually top out.



Neighboring plan

Search Decisions



Method	Decision on the same planning problem
Hill climbing	Scan all neighbors; accept improvement or an allowed sideways move.
Random restarts	Begin another hill-climbing episode with a new complete plan.
Simulated annealing	Propose one neighbor; sometimes accept a lower fitness.
Local beam search	Pool all parents' neighbors; retain the best distinct successors.
Genetic algorithm	Combine whole placements (genes) from parents; sometimes replace one placement.

Student Work and Evidence

Supplied

Shapes, domains, neighbors, simulation, fitness, search helpers, random baseline, genetic loop, and replay tools.

Six required functions

Hill climbing, random restarts, annealing, beam search, crossover, and mutation.

Independent evidence

Hand-worked traces; tests with known answers; reproducible runs; replay observations.

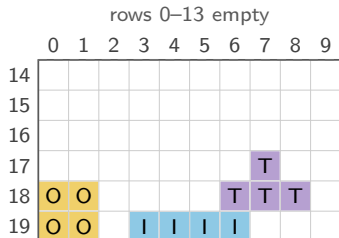
Experiments and report

Compare five levels and three seeds under a 10,000-evaluation limit. Report actual evaluations and fitness components.

Report: at most eight pages, including traces, figures, tables, and references.

Understanding Check

Three-piece example: O, I, T; $N = 3$.

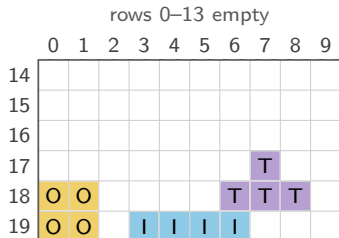


Plan B after all three drops

1. Is changing I from (0, 3) to (1, 9) one neighbor step?
2. How many holes are in this board?
3. Compared with Plan A, why does Plan B have higher fitness?

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Answers

Only I's placement changes, so yes.

Two holes, below the right side of T.

P and L tie; Q decreases from 82 to 32.

Thank You

Any questions?